



New data on temperature optimum and temperature changes in energy crop digesters

H. Lindorfer^{b,*}, R. Waltenberger^a, K. Köllner^a, R. Braun^a, R. Kirchmayr^a

^a BOKU – University of Natural Resources and Applied Life Sciences, Vienna, Department for Agrobiotechnology, IFA-Tulln, Institute for Environmental Biotechnology, Konrad Lorenz Str. 20, 3430 Tulln, Austria

^b Schaumann BioEnergy, An der Mühlenau 4, 25421 Pinneberg, Germany

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Abstract

As a result of self-heating in anaerobic digesters when using energy crops in the feedstock, the influence of temperature on the digestion process came back into focus. The aim of this study was to investigate the impact of such temperature increases on process stability. Furthermore, different strategies for the transition from mesophilic to thermophilic conditions and the resulting methane yields at different temperature levels were evaluated. Two main effects were identified with different bio-slurries from agricultural biogas plants: (1) a failure of methane production connected to changes in the microbial community; and (2), a slow but continuous accumulation of propionic acid, though without an immediate effect on methane production. All strategies for increasing the operating temperature showed negative effects on digester performance, some with serious economic consequences for the operator. It was shown that methane yields at different temperature levels in the mesophilic and sub-thermophilic ranges are similar.

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1. Introduction

Many investigations have already been carried out on the advantages and disadvantages of thermophilic versus mesophilic conditions in sewage sludge digestion (e.g. van Lier et al., 1993, 1996; Speece, 1996; Gallert and Winter, 1997; Záborská et al., 2000; Angelidaki et al., 2004; Kim et al., 2006).

In most studies, temperature optima were defined either in the mesophilic range between 35 and 40 °C or in the thermophilic range between 50 and 65 °C (e.g. Gerardi, 2003; Bischofsberger et al., 2004), and thus, most current biogas plants operate either in the one or the other temperature range. In contrast, there are almost no data available

about process temperatures between 40 and 50 °C, other than a general statement that this range is unfavourable for anaerobic digestion (Gerardi, 2003).

However, the temperature range between 40 and 50 °C is becoming more important again owing to the effects of self-heating in mesophilic energy-crop digestion. As a consequence of exothermic carbohydrate degradation, the high energy density in the substrates used and the high loading rates, self-heating effects have led to an increase in process temperatures from 35–39 °C to 42–49 °C, depending on the individual reactor design (Daverio et al., 2003; Lindorfer et al., 2006; Braun, 2007). In 2005, already 20 out of the 41 biogas plants in Austria surveyed were being operated at temperatures between 40 and 50 °C, mainly due to this effect (Laaber et al., 2005; Braun, 2007). The velocity of temperature increase depends on the individual plant situation, but rates between 0.15 and 0.45 °C d⁻¹ were observed. The only possibilities available to prevent these temperature increases are to change the feedstock, to

* Corresponding author. Tel.: +49 41012185412; fax: +49 41012185499.

E-mail address: harald.lindorfer@schaumann-bioenergy.eu (H. Lindorfer).

reduce the organic loading rate or to install a cooling system similar to that used for aerated systems.

In several cases the unwanted increase in temperature caused severe disturbances in biogas production (Lindorfer et al., 2006). Furthermore, laboratory-scale experiments of other authors showed complete failures in methane production for several days (Bolzonella et al., 2006). For the plant owners, it is important to assess the impact of the self-induced heating of their digesters, since a breakdown of the kind described above would cause severe economic damage.

In this study the effects of such temperature increases on digester performance were investigated in laboratory and pilot-scale. Furthermore, different strategies for the transition from 38 to 45 and 48 °C were evaluated to select the most advantageous. Even though some literature on the temperature susceptibility of anaerobic populations can be found (van Lier et al., 1990; Ahn and Forster, 2002; El Mashad et al., 2004), the currently observed effects after self-induced temperature increases in mesophilic energy crop digesters do not fit the literature data and cannot yet be predicted. Some of the studies in this field were conducted on the effects of temperature drops (e.g. Massé and Masse, 2001) and about the impact of temperature increases on thermophilic processes (van Lier et al., 1993; Ahring et al., 2001; Iranpour et al., 2005). Temperature increases in mesophilic systems were only studied by a few authors (e.g. Lescure et al., 1988).

The effects during the transition from mesophilic to thermophilic conditions were investigated by van Lier et al. (1993), Ahn and Forster (2002) and Bolzonella et al. (2006). Bousková et al. (2005) presented different strategies for the transition in sewage sludge digestion. Still, all these studies were conducted in different digestion systems (e.g. UASB, CSTR or 'semi dry systems') and with different substrates (e.g. sewage sludge, industrial wastewater or organic municipal wastes) severe process disturbances were observed by all authors when temperatures exceeded 45 °C. The parameter mostly affected by the temperature

fluctuations in almost all studies was propionate degradation. Drops in methane production, and even a complete failure of methanogenic activities for 20 or more days were also observed.

The available information from literature is not enough to calculate the impacts on process stability or to predict the possible consequences of a temperature increase in praxis scale plants. As the self-heating effect is often the driving force for a temperature change, it is necessary to investigate strategies to prevent process disturbances. Furthermore, the advantages or disadvantages of different temperature levels, for example on ammonia inhibition, digester efficiency or heating requirements need to be evaluated as well. Thus it is about time that more attention is paid to the sub-thermophilic temperature range between 40 and 50 °C.

2. Methods

2.1. Pilot-scale experiments

The pilot plant in which parts of this study were realized situated on the site of a full-scale agricultural biogas plant was a two stage continuous stirred tank reactor (CSTR) system with a first (main) digester volume of 2 m³ and a second digester step of 0.7 m³ (Fig. 1). All data of the present study were recorded in the first digester. The substrates used were either silage from maize grains or whole-crop maize silage and pig manure. Feedstock data are shown in Table 1. The chosen solid feed dosage was provided once daily through a vertical funnel. All input weights were recorded, but the amount of manure was recorded volumetrically. Temperature was measured inside the digester 40 cm above digester base at 15 cm from the digester wall. The reference process temperature at the start of all experiments was 38 °C. Biogas quantity was measured by conventional mechanical gas counters (Krom Schröder BK G2.5). Gas quality was analysed online, using Awite equip-

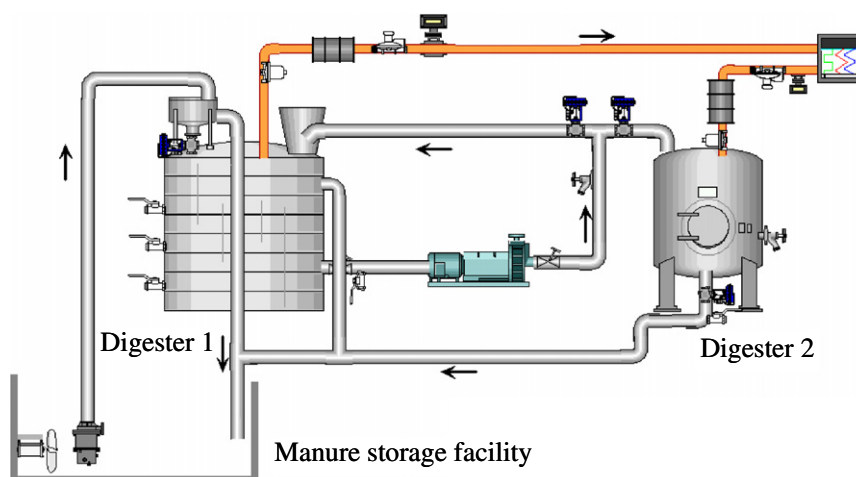


Fig. 1. Schematic of the pilot-scale experiments.

Table 1
Chemical data of inoculi and substrates used in the study

	Bio-slurries			Substrates		
	BS 1	BS 2	BS 3	WCS	MGS	Manure
pH	7.67	7.63	7.65	–	–	–
COD [g l ⁻¹]	69.6	119.4	112.9	464.2	936.1	84.5
TS [%]	6.19	9.44	10.14	36.51	68.22	5.74
VS [%]	4.55	7.31	7.98	35.05	66.51	4.16
TKN [g l ⁻¹]	3.19	8.19	9.09	3.25	13.91	5.13
TAN [g l ⁻¹]	1.46	4.39	5.00	–	–	3.65
FAN [mg l ⁻¹]	61	235	281	–	–	–
Acetic acid [mg l ⁻¹]	46	860	771	–	–	–
Propionic acid [mg l ⁻¹]	42	186	168	–	–	–
Total VFA [mg l ⁻¹]	103	1051	979	–	–	–

MGS – Maize grain silage, WCS – Whole crop silage.

ment (Ser. 3) based on infrared detectors (CH₄ and CO₂) and electrochemical cells (O₂ and H₂S).

2.2. Laboratory-scale experiments

The laboratory digesters consisted of three glass reactors ($V_{\text{tot}} = 2.0$ l) each containing 1.5 l of bio-slurry (BS). In experiments 1–4 the substrate was silage from maize grains, in experiments 5–7 it was whole-crop maize silage. The digester systems were placed in an incubation chamber or in incubators at different temperatures. Control tests were carried out at 35 °C. Feeding was once daily. The digesters were agitated manually 2–4 times daily. Biogas was collected and passed through a 4 M NaOH solution to remove CO₂ from the gas. Methane production was measured with micro-gascouters (Fa. Ritter, Germany).

2.3. Feedstock and inoculum

Three different bio-slurries (BS 1–3) were used in this study. They were all taken from agricultural digesters using a feedstock of manure and energy crops. The specific data of the bio-slurries are shown in Table 1. BS 1 represented a bio-slurry with low ammonia content, low organic loading rate (OLR) in former processing and a sub-mesophilic thermal history with temperatures never higher than 37 °C. BS 2 and BS 3 were taken out of the first and the second digester of the same biogas plant. The particularities of BS 2 were a medium level ammonia content, high OLR and a mesophilic history with a temperature average at 39 °C, but maxima up to 41 °C due to self-heating effects. BS 3 had the highest ammonia content (5.5 g l⁻¹), a high OLR and the same temperature history as BS 2. While laboratory fed-batch experiments were conducted with BS 1 and BS 2, in the pilot-scale experiments BS 2 and BS 3 were used. In the majority of the experiments, the temperature was increased in one step from 35 to 48 or 50 °C over 24 h in laboratory-scale studies and from 38 to 45 or 48 °C over 36 h in the pilot-scale ones. In other experiments, temperature was increased slowly at a rate of 1 °C

per week, during which the influence of short-term temperature shocks before the final increase were investigated.

2.4. Sampling and chemical analysis

Samples of the digester contents were taken weekly after intensive mixing. The pilot-scale plant was sampled at the vertical part of a recurrent loop, as recommended by Holm-Nielsen et al. (2006). The sampling procedure always consisted of discarding the old contents of the sampling tube, then drawing off a sample of 10 l in several steps into a bucket, and then taking an aliquot of 0.5 l out of the bucket after mixing. The laboratory digesters were sampled by means of sampling tubes in the lower part of the digesters with each sample being approx. 30 ml. All samples were cooled to 4 °C immediately after sampling. Chemical analysis was performed within 48 h. Volatile fatty acids (VFA) were analysed using high performance liquid chromatography (HPLC System, Agilent 1100 Series, RI-Detector). Total nitrogen (TKN, Kjeldatherm, Fa. Gerhardt GmbH), total ammonia nitrogen (TAN, Vapodest 5, Gerhardt GmbH), pH (340i, WTW), total solids (TS) and volatile solids (VS) were determined using standard methods (APHA 1998). Free ammonia nitrogen (FAN) was calculated using the Hobiger's formula (1996)

$$c(\text{FAN}) = \frac{M(\text{NH}_3)}{M(\text{NH}_4)} \times \frac{c(\text{TAN}) \times 1000}{\left(1 + 10^{\left(\frac{(0.0925 + 2728.795)}{(T + 273.15)}\right)}\right) - \text{pH}} \quad (1)$$

2.5. Calculation of process parameters

The OLR is the quotient of daily mass of volatile solids (VS) in the feedstock and the volume of the liquid digester contents (bio-slurries), characterised by the dimension [kg VS m⁻³ d⁻¹]. The hydraulic retention time (HRT) was estimated according to Helffrich (2005) as a quotient of the daily feedstock mass fresh matter (FM) and the total digester volume, since the volumetric calculation is not useful in solid substrate digestion. The volume related biogas productivity is the quotient of the volume of daily biogas production and total digester volume [dimensions of m³ (m³ d)⁻¹].

3. Results and discussion

3.1. Effects of temperature increase on different bio-slurries

The effects of temperature increase on digester performance observed could be separated into two main categories: a drop of methane production immediately after the temperature change (effect 1) and continuous accumulation of propionic acid, maintaining methane production on a high level (effect 2). While the first effect only occurred in the experiments using BS 1, the accumulation of propionic acid was observed in all experiments. The temperature changes in laboratory-scale experiments were carried out

during 12–24 h, in pilot-scale experiments the changes were finalized after 24–48 h. In the following, the results are subdivided according to the different effects observed.

3.2. Effect 1: Drop of methane production

In all three experiments with BS 1, a breakdown of the methane production was observed at temperatures exceeding 45 °C. Fig. 2 shows methane production after a temper-

ature change from 35 to 48 °C. The first visible effect was a temporary increase in methane production. This is to some extent the effect of degassing of volatile components, but the major part is caused by increased microbial activity at higher temperatures (van Lier et al., 1990). After the transient increase, the methane production decreased rapidly to 10% of the original level. At the same time, VFA concentrations increased from 0.15 (0.13) to 11.88 g l⁻¹ (11.12 g l⁻¹). The time needed for methane production and VFA levels to recover was approx. 20 days in both experiments with same setup. This is twice the time as in the study of Bolzonella et al. (2006), in which gas production had already recovered after 10 days, though process stability was not reached within less than 30 days.

After the drop in methane production, feeding was suspended until gas production recovered. In this period, VFA concentrations decreased to 2.88 g l⁻¹. However, if feeding was continued at 48 °C at the original OLR, methane production and acetic acid concentrations showed good performance, but propionic acid started to accumulate continuously. A screening of the microbial population showed severe shifts in methanogenic community structure that accompanied the drop in methane production (data not presented). This matches the results of van Lier et al. (1996), who found that thermophilic bacteria are initially absent before transition from mesophilic to thermophilic conditions. A detailed analysis of the population was not carried out in this study, but the results suggest that, in BS 1, a bacterial population change took place during the period of low productivity. While most positions in the anaerobic food chain were reoccupied during that period, the propionate degraders seemed to be more affected.

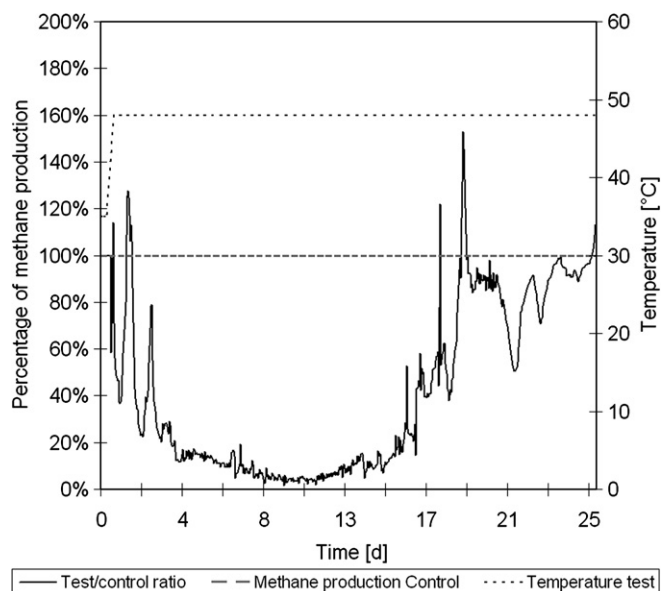


Fig. 2. Methane production ratio from BS 1 during the change from 35 to 48 °C compared with that of the control operated at 35 °C.

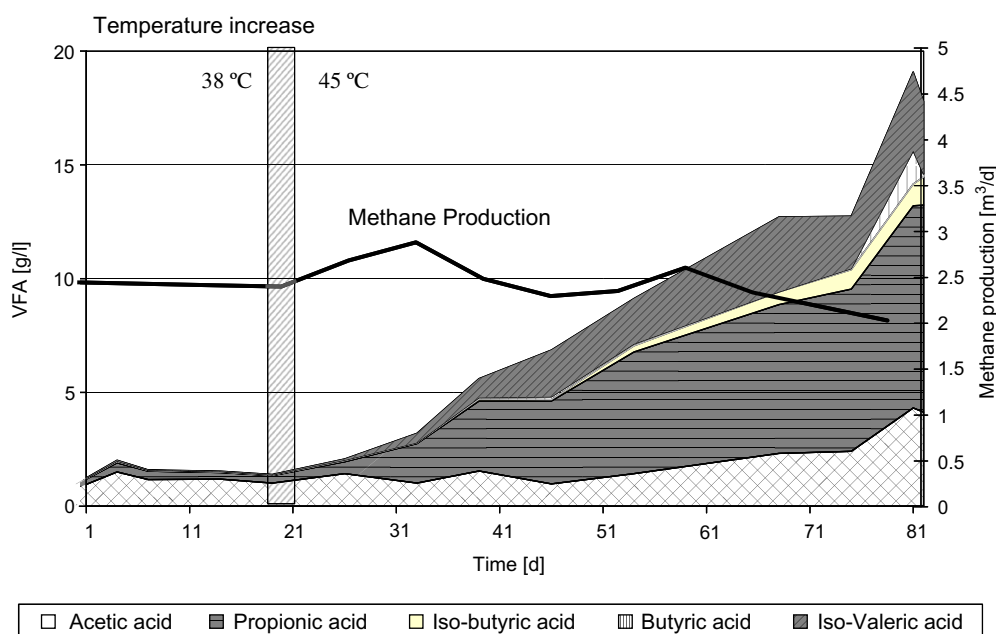


Fig. 3. Methane production (NTP) and accumulation of volatile fatty acids in BS 3 during a temperature increase from 38 to 45 °C in a pilot-scale experiment.

3.3. Effect 2: Accumulation of propionic acid

Temperature changes with the BS 2 and BS 3 showed different behaviour. Although a drop in methane production when the temperature exceeded 42 °C had been observed one year before in the full-scale plant from which the bio-slurries were taken (Lindorfer et al., 2006), no similar effects occurred during the experiments of this study, even when the temperature was increased up to 55 °C. Methane production stayed at the same level, unimpaired by the temperature changes (Fig. 3). However, after a temporary increase in the levels of acetic and propionic acids, a slow but continuous accumulation of propionic, iso-butyric and iso-valeric acids started. Retaining the same OLR and temperature, this accumulation continued until process breakdown. The maximum concentrations reached were 9.4 g l⁻¹ of propionic acid, 1.3 g l⁻¹ of iso-butyric acid and 3.3 g l⁻¹ of iso-valeric acid. The total concentration of VFA at this point was 19.2 g l⁻¹. Even after three weeks without feeding, the concentration only decreased by 6.1%, after another two weeks by 14.6%.

These observations correspond to data of other authors, who found that degradation of propionic acid is the most sensitive step during transition from mesophilic to thermophilic conditions (van Lier et al., 1990,1993; Lescure et al., 1988). Only Ahn and Forster (2002) observed a rapid decrease in propionic acid after a return to mesophilic temperatures.

While with BS 3 the process disturbances mentioned above were already visible after a temperature increase from 38 to 45 °C (Fig. 3), in the experiments with BS 2, the first negative effects were observed between 45 and 48 °C (Fig. 7).

Considering the small chemical differences between BS 2 and BS 3, it is the FAN concentration that seems to be the decisive parameter. As a consequence of the higher temperatures the FAN to TAN ratio increased, leading to higher FAN concentrations in the bio-slurries (Eq. (1)). Since the temperature was increased over a very short period, the FAN to TAN ratio in the bio-slurries changed rapidly as well. Although the population was adapted to high ammonia levels, it was harmed by the fast increase, as the acclimatization to higher ammonia levels needs several months (Angelidaki and Ahring, 1994; Gallert and Winter, 1997; Gerardi, 2003; Calli et al., 2005). In Fig. 4, the changing FAN concentration in BS 2 during the period of temperature increase is shown. As can be seen, the accumulation of propionic acid started directly after the main FAN peak. However, only the propionate degrading bacteria were severely affected by the FAN peak. Other organisms involved in the anaerobic food chain seemed not to be influenced.

As the FAN level is strongly correlated with the fluctuating pH values in the digester related to the feeding intervals, TAN concentration is the parameter more often applied to assess the critical nitrogen level. According to the present study, the critical TAN concentration at

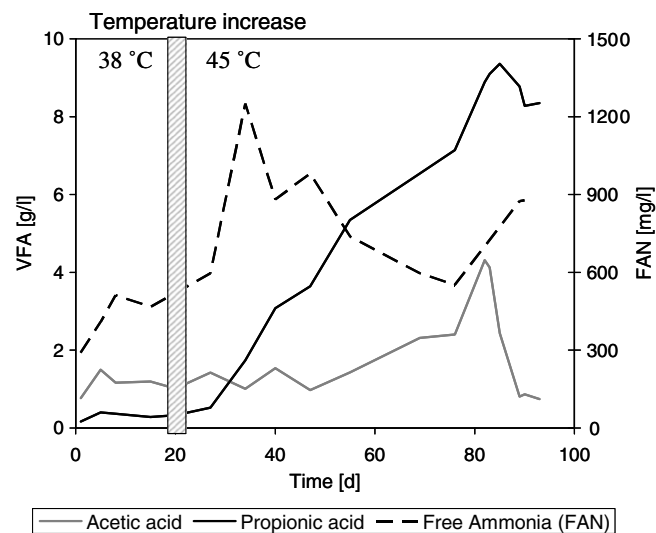


Fig. 4. Free ammonia nitrogen and VFA dynamics in BS 3 during the change from 35 to 45 °C in a pilot-scale experiment.

45 °C seemed to be at 5.5 g l⁻¹; at 4.0 g l⁻¹ no disturbance was observed. Angelidaki and Ahring (1994) recorded the critical TAN level in anaerobic digestion at 45 °C at 6 g l⁻¹.

3.4. Strategies for the transition from mesophilic to thermophilic processing

First studies of strategies for a controlled temperature increase in digesters were carried out by Bousková et al. (2005), who compared a one step with a multi-step strategy. In summary they recommended the former, as the cumulative biogas losses were lower. Some experiences from full-scale plants indicate that a temperature increasing rate of 1 °C per week effects no process disturbances (Braun 2007). In this study the experiments had to be divided in two categories as the temperature increase showed different reactions within the bio-slurries used. In the experiments, BS 1 represented a bio-slurry which was expected to undergo a population change, while BS 2 represented a bio-slurry in which there were already sufficient thermophilic bacteria to tolerate the transition from mesophilic to sub-thermophilic conditions.

Three strategies were tested to evaluate the transition from mesophilic to thermophilic process in the temperature tolerant bio-slurry (BS 2): (1) a temperature increase in a single step from 35 to 48 °C; (2) a slow, multiple step increase (1 °C/week); and (3) a single step increase after short-period (24 h) temperature pulses. The temperature increase had no significant influence on methane production in any of the experiments. However, all experiments showed a slow but continuous accumulation of propionic acid (Fig. 5a–c). As the increase continued, the feed supply had to be reduced after some weeks. The strategy of delivering short-period temperature pulses before the final increase had just a very small habituation effect on the microbial community.

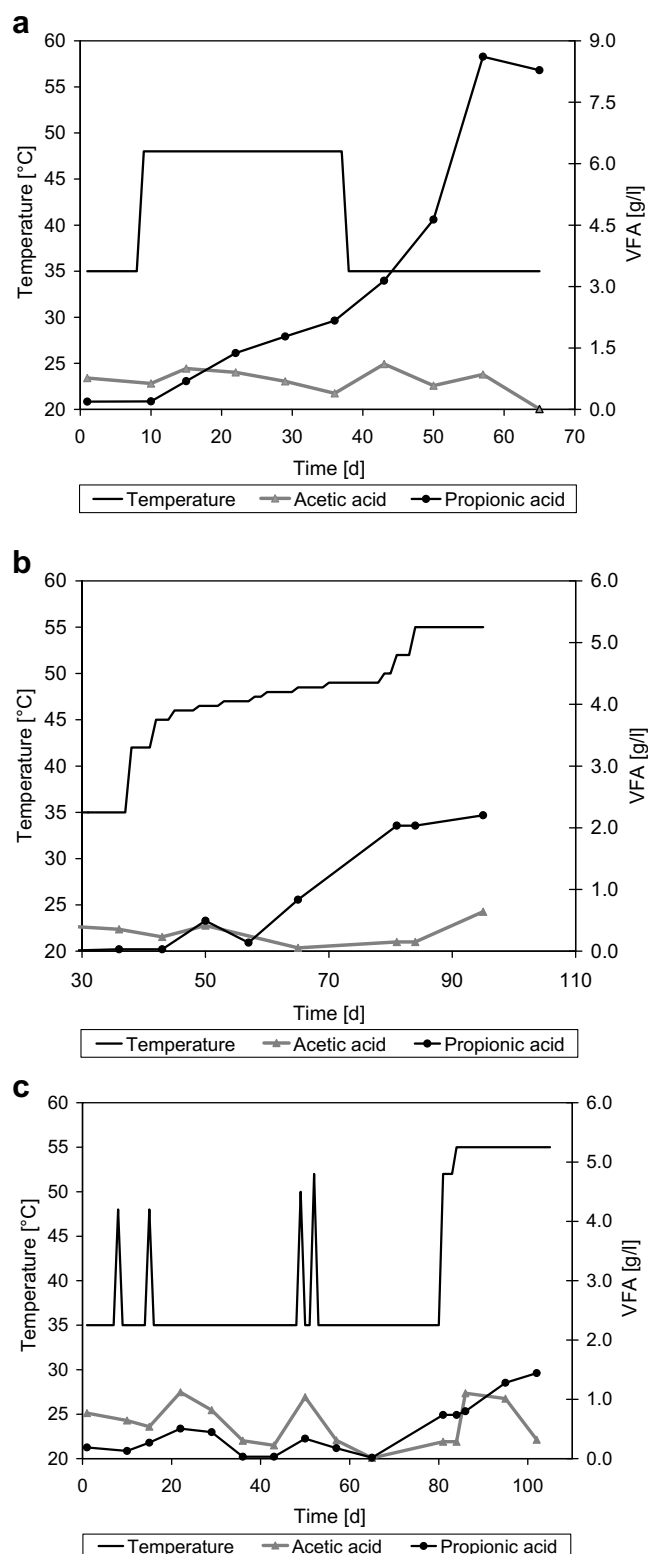


Fig. 5. (a–c) Strategies for the transition from mesophilic to thermophilic conditions with temperature tolerant bio-slurries: (a) temperature increase in a single step; (b) slow, multiple step increases; (c) single step increase after short temperature pulses (Note: VFA scale differences!).

With BS 1, two strategies were tested: (1) a temperature increase in a single step from 38 to 48 °C; and (2), a single

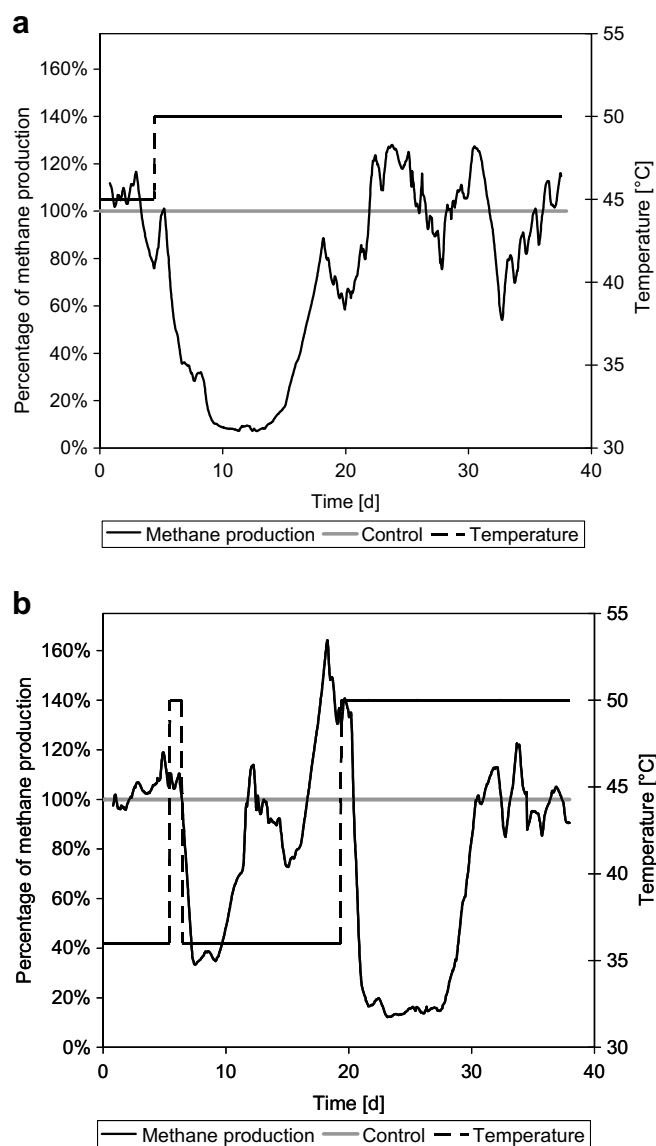


Fig. 6. (a, b) Strategies for the transition from mesophilic to thermophilic conditions with temperature intolerant bio-slurries: (a) temperature increase in a single step; (b) single step increase after a short temperature pulse. Data shown as percentage of methane production related to the control operated at constant temperature.

step increase after a short-period (24 h) temperature pulse (Fig. 6a and b). Both experiments showed an immediate drop in methane production after temperature increase. Owing to the drop of methane production, feeding was suspended until full recovery. The recovery period was 20 days in (1) and 9.6 days in (2). However, in the latter case, a drop in methane production for some five days already occurred after the first 24 h temperature pulse. VFA increased immediately after the temperature change in both experiments and then, during the recovery period, the content of VFA decreased to low levels. Later operation took place with the original VFA levels. In summary, the second strategy was still the better. It therefore appears that, in BS 1, the strategy of applying short temperature pulses before applying the final increase had a habituation effect on the

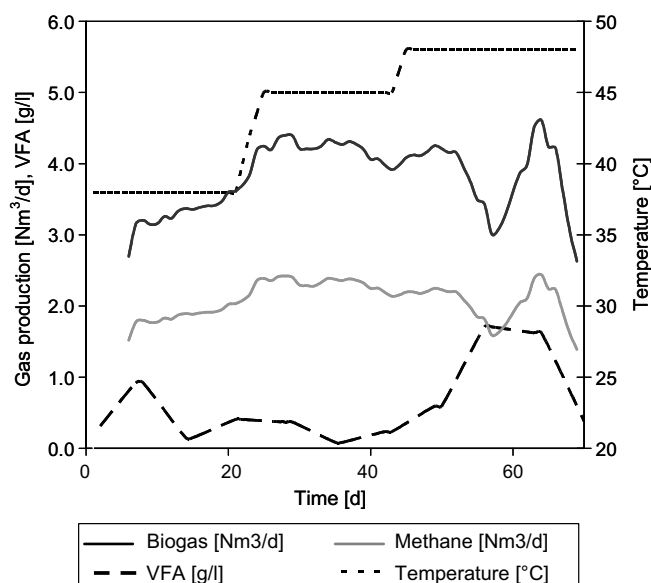


Fig. 7. Methane yields during transition from mesophilic to thermophilic conditions (BS 2). $\text{N m}^3 = \text{m}^3$ (NTP).

microbial community. It can be assumed that an enhancement of the habituation effect would be possible, with the implementation of more and shorter pulses. However, applying that strategy in full-scale plants seems to be difficult because of their limited possibilities to carry out short-period temperature changes.

Experiences with full-scale plants show that negative effects on process performance can be reduced and a significant reduction of propionic acid levels can be achieved by the addition of active biomass. This was also recommended by van Lier et al. (1990). In full-scale operation this can be organised easily by recirculation of bio-slurry from the second digestion step or a covered effluent storage into the main digester. Furthermore, a severe accumulation of VFA during the temperature change and the following process disturbances can be avoided by suspending, or at least reducing, substrate feeding in time before increasing temperature, as temperature sensitivity increases with loading rate (Speece 1996).

In a full-scale case study, it was shown that a breakdown of microbial activity due to self-induced temperature increases could be avoided by starting-up the plant already at a sub-thermophilic temperature level. The following temperature increase to 49 °C over a few weeks by self-heating then took place without any disturbances (data not shown).

3.5. Temperature optimum in the anaerobic digestion of energy crops

As already mentioned, most current biogas plants operate in the temperature ranges of 35–40 °C or 50–65 °C. The temperature range between 40 and 50 °C was avoided as being of lower efficiency.

This widespread assumption was disproved in the present study. In Fig. 7, methane and biogas production from BS 2 over a three-month period (pilot-scale), including two temperature steps from 38 to 45 °C and to 48 °C are shown. The OLR was maintained on the same level until the VFA increased rapidly at day 52. At that point feeding was suspended. As can be seen, methane production increased slightly at the temperature of 45 °C; methane yield increased at the same time approx. 8%. After a further temperature step to 48 °C, methane production stayed almost at the same level until feeding was stopped due to high VFA concentrations. In a similar experiment with BS 3, which had a higher TAN level, methane yield at 45 °C was 7% lower than at 38 °C (Table 2). As already discussed in the previous section, ammonia inhibition seemed to be the reason. Temperature and OLR related methane yields of all experiments are presented in Table 2. Note that laboratory-scale data represent an average over two weeks before and after temperature changes. In these short periods, stable conditions could not always be achieved.

In summary, it can be assumed that the wide biodiversity in the anaerobic bacterial populations of full-scale plants at mesophilic and sub-thermophilic temperatures have obviously prevented the activity losses described in

Table 2
Methane yields (Y_{CH_4}) in different experimental setups

	Laboratory-scale ^b	Pilot-scale	Praxis scale
BS 1			
OLR [kg VS m^{-3}]	2.2		
Y_{CH_4} (35 °C) [$\text{N m}^3 \text{ kg VS}^{-1}$]	0.40 (± 0.12)	–	–
Y_{CH_4} (48 °C) [$\text{N m}^3 \text{ kg VS}^{-1}$]	0.43 (± 0.02)	–	–
Y_{CH_4} (50 °C) [$\text{N m}^3 \text{ kg VS}^{-1}$]	0.30 (± 0.10)	–	–
Y_{CH_4} (55 °C) [$\text{N m}^3 \text{ kg VS}^{-1}$]	0.28 (± 0.09)	–	–
BS 2			
OLR [kg VS m^{-3}]	2.3	2.9	4.5
Y_{CH_4} (38 °C) [$\text{N m}^3 \text{ kg VS}^{-1}$]	0.34 (± 0.05) ^a	0.35 (± 0.06)	0.37 (± 0.05)
Y_{CH_4} (45 °C) [$\text{N m}^3 \text{ kg VS}^{-1}$]	–	0.38 (± 0.04)	
Y_{CH_4} (48 °C) [$\text{N m}^3 \text{ kg VS}^{-1}$]	0.22 (± 0.04) ^c	0.36 (± 0.01)	
BS 3			
OLR [kg VS m^{-3}]	–	4.4	4.3
Y_{CH_4} (38 °C) [$\text{N m}^3 \text{ kg VS}^{-1}$]	–	0.31 (± 0.04)	0.32 (± 0.03)
Y_{CH_4} (45 °C) [$\text{N m}^3 \text{ kg VS}^{-1}$]	–	0.29 (± 0.04)	–

Average values during 14 days in laboratory-scale (note that stable performance was not always achieved), 21 days in pilot-scale and 90 days in praxis scale. $\text{N m}^3 = \text{m}^3$ (NTP).

^a 35 °C.

^b Stable performance was not always achieved.

^c Technical problems.

single strain experiments. The results of the present study show that there are no general temperature optima in anaerobic digestion. The individual optimum depends on other, plant specific parameters, mainly self-heating potential, nitrogen content of the substrate mix or the possibility to sell the heat. For hygienic reasons, plants accepting slaughterhouse wastes in the feedstock might well favour temperatures higher than 55 °C. Other advantages of higher temperatures are the faster metabolism and conversion rate (van Lier et al., 1996), and a lower viscosity of the bio-slurries which might reduce the stirring demand. Disadvantages of the thermophilic process are a lower process stability (Kim et al., 2002) as a result of the lower biodiversity and a significantly higher demand for heating energy (Zupancic and Ros, 2003). Further possible problems are the effect of solid–liquid separation, which can lead to the formation of layers, and the risk of ammonia inhibition. Braun (1982) suggested that the temperature optimum of a plant is always a compromise between the temperature optimum of the microbial population and the energy balance of the whole plant.

One general aim should be the avoidance of temperature fluctuations as they influence process stability in a negative way as a result of the temperature sensitivity of methanogenic species (Speece, 1996; Iranpour et al., 2005). Therefore, in plants with self-heating effects, the optimum temperature will be the maximum level of self-induced heating. In colder seasons, the temperature difference due to heat radiation can be compensated for with a minimum additional heating. For most other plants, the operators can decide individually between 35 and 65 °C considering the factors mentioned above.

Special attention has to be paid to biogas plants with high nitrogen loading. As is shown by the results presented above, the critical temperature of ammonia-rich bio-slurries is lower than 45 °C. This is no problem for the plants using slaughterhouse wastes or other wastes rich in fat and proteins, but for digesters using mainly cereal grains as feedstock or other carbohydrate and nitrogen rich substrates. In the latter cases, the self-heating might lead to a temperature increase to levels higher than 45 °C. The only way to preserve mesophilic conditions besides changing the feedstock itself, which is not possible in most cases, requires the installation of a cooling facility, as is the case for the biogas plant from which BS 2 and 3 were taken.

4. Conclusions

The results of this study show that there are no general temperature optima in anaerobic digestion as was originally concluded on the basis of single strain experiments. In the range from 35 to 50 °C, methane yields are similar. The individual temperature optimum of a biogas plant depends on many other parameters such as nitrogen load, hygiene requirements and plant design.

The self-induced temperature increase in mesophilic digesters to sub-thermophilic levels (40–50 °C) can cause

severe process disturbances. Whatever strategy is chosen to increase temperature, whether one step or multiple step, or after applying short temperature pulses, some negative impacts (e.g. on methane output or propionic acid formation) are rarely avoidable. However, the strategy of habituation of the microbial community to higher temperatures by short temperature pulses could be the best way to increase temperature in biogas digesters. Addition of new microbial mass, for example by recirculation of bio-slurry from other digester steps, is another possibility to reduce negative impacts.

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